



PYTHON

USING APIS WITH REQUESTS



SESSION: WHAT IS AN API? TALKING TO THE INTERNET WITH PYTHON

HAVE YOU USED ZOMATO OR SWIGGY?

CONNECTING TO APPS YOU ALREADY KNOW AND LOVE!

Think about when you order food at home. You open an app, pick your favorite biryani or pizza, and tap "Order"! But how does the restaurant know what you want? There's something special happening behind the scenes - and Python can do the same thing!

→ DO YOU ORDER FOOD WITH ZOMATO OR SWIGGY?

→ HOW DO YOU TELL THE RESTAURANT WHAT YOU WANT?



PYTHON CAN
TALK TO APPS
JUST LIKE YOU
DO!

LET'S EXPLORE!

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PYTHON TALKS TO THE INTERNET!



- Python can talk to internet apps! We send messages and get answers back. It's like having a super smart helper that can ask questions anywhere online!
- Let's find out how it works! You'll be amazed at what Python can do when it connects to the internet. Ready to explore? 🚀





WHAT IS AN API?



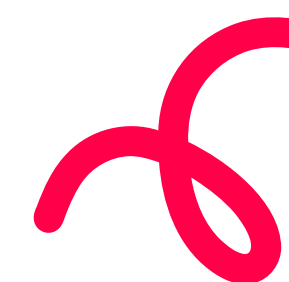
**API = WAITER IN A
RESTAURANT!**

→ THE RESTAURANT WAITER ANALOGY

An API is like a helpful waiter! When you want something from a restaurant kitchen, you don't go inside yourself. You tell the waiter what you want, and the waiter brings it back. Python uses APIs the same way to get information from the internet!

You tell the waiter your order. The waiter takes your order to the kitchen. The kitchen makes the food. The waiter brings food back to you. That's exactly how APIs work!

PYTHON IS YOU, API IS THE WAITER!





REQUEST AND RESPONSE



REQUEST
"BROTHER, ONE TEA
PLEASE!" ☕



RESPONSE
TEA IS SERVED! ☕



ERROR 404
NO TEA TODAY

Think of it like ordering chai:

- You say: "Brother, give me one cutting chai (half tea)!"

" = This is your REQUEST

- he tea seller gives you tea = This is the RESPONSE

- Status Code 200 = Success! Tea ready!

What if something goes wrong?

- There is no milk today = ERROR RESPONSE

- Status Code 404 = Not Found!

- URL = The tea seller's address where you go to order

Request is your question, Response is the answer!





JSON DATA FORMAT



JSON is like your Aadhaar card data!

It stores information neatly:

- Name
- Date of birth
- City
- ID number

JSON is what computers share with each other!



DATA ORGANIZED
LIKE AN AADHAAR
CARD!

Example JSON data:

```
{'name': 'Priya Sharma',  
'dob': '2015-04-12',  
'city': 'Bengaluru',  
'aadhaar': '1234-5678-9012'}
```

It looks just like a Python dictionary!

REQUESTS LIBRARY

YOUR JIO SIM!

PYTHON NEEDS REQUESTS LIKE YOU NEED JIO SIM TO MAKE CALLS!

The requests library is Python's special tool to talk to the internet. Just like your phone needs a JIO SIM card to call your friends, Python needs the requests library to call websites and APIs!

→ **IMPORT AT TOP: IMPORT REQUESTS**

→ **ASK FOR DATA: REQUESTS.GET(URL)**



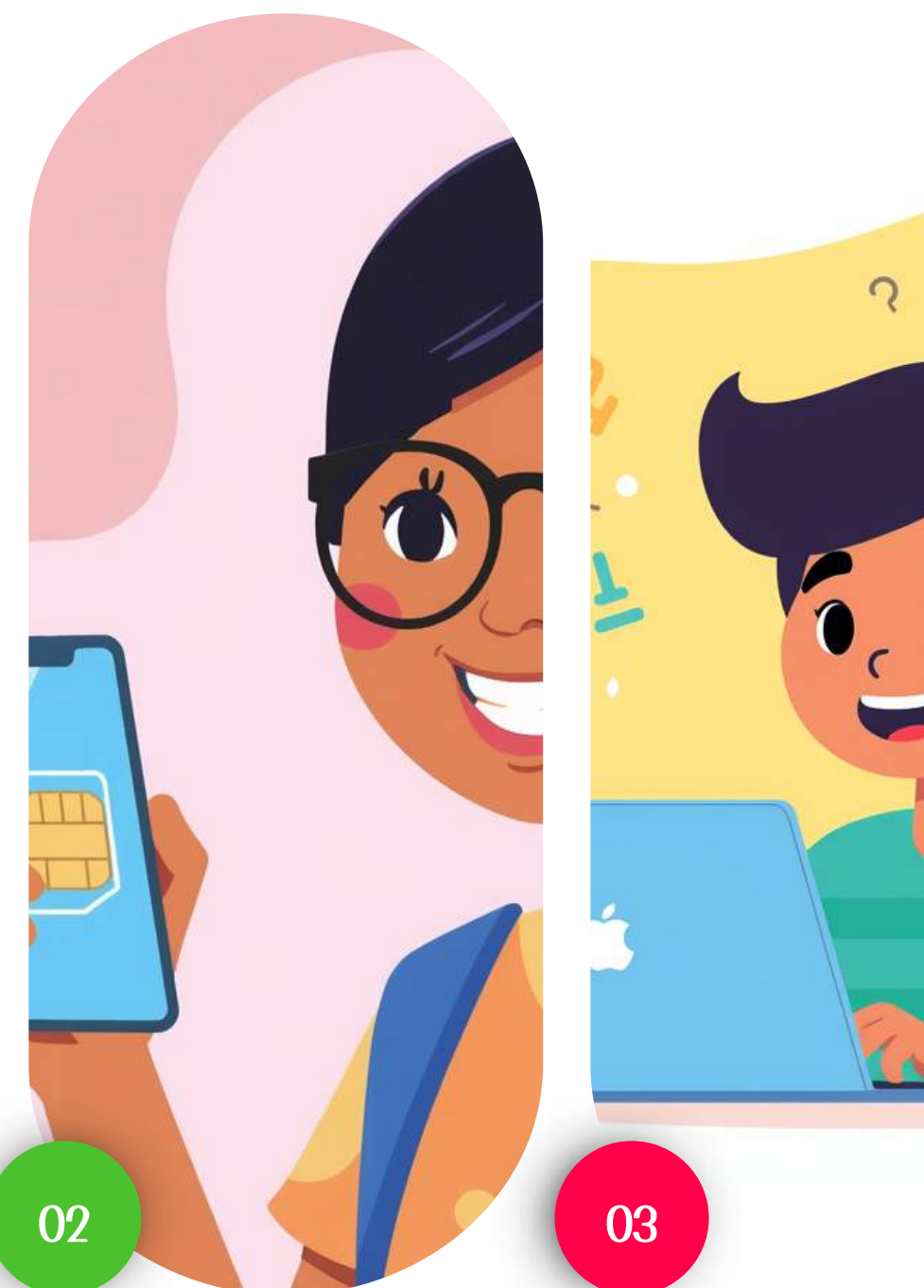
**INSTALL IT FIRST:
PIP INSTALL
REQUESTS**

LET'S LEARN!

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HOW APIS WORK THE ZOMATO WAITER ANALOGY



YOU = Python Program
You decide what food you want to order.
Your Python code decides what data to ask
for!



WAITER = API
The waiter takes your order to the kitchen.
The API carries your request to the server!



KITCHEN → FOOD = Server → Response
Kitchen makes food and sends it back.
Server processes request and sends data
back!



LET'S WRITE SOME CODE!



TIME TO BECOME A PYTHON CODER!

Get ready for some fun coding! We're going to write a Python program that asks the internet for a joke. Yes, really! Python can fetch jokes from the internet all by itself. Let's see how it works step by step!

READY? LET'S GO!



WE WILL USE PYTHON
TO GET A JOKE FROM
THE INTERNET!

Here is the first line of code we will write:

```
import requests
```

This tells Python to use the requests library so we can talk to the internet!

WELL DONE, YOU ARE CODING! 🎉





THE API URL

FINDING THE ADDRESS!

Every API has a special address called a URL. This is where Python sends its request to get data. Think of it like a phone number - you need the right number to call the right person!

URL

```
URL = "HTTPS://OFFICIAL-JOKE-  
APIAPPSPOT.COM/RANDOM_JOKE"
```

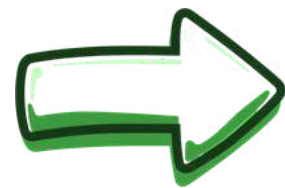
10/21

PYTHON ASKS THIS ADDRESS FOR A JOKE - JUST LIKE DIALING A FRIEND!





SENDING THE REQUEST!



Here's the magic line of code:

```
response = requests.get(url)
```

This tells Python to send your request to the API address!



`requests.get()` sends our request to the API server. The 'response' is what the API sends back to us - like getting a reply to your message! 



#

READING THE JSON RESPONSE

Once we get the response from the API, we need to read the data! We use a special command to turn the response into something Python can understand.

```
Code: data = response.json()
```

This reads the joke data into Python as a dictionary!

SETUP

THE FIRST PART OF THE JOKE - THE QUESTION OR BEGINNING

PUNCHLINE

THE FUNNY ANSWER THAT MAKES YOU LAUGH!





STEP 1: IMPORT REQUESTS

IMPORT REQUESTS

We start by importing the requests library.
This is always the first line in our code when
we want Python to talk to the internet!

TYPE THIS LINE AT THE TOP OF YOUR CODE!

LIVE CODING





DEFINE THE URL ADDRESS

STEP 2

The URL is like a phone number for the internet! It tells Python exactly where to go to ask for our joke. Every API has its own special address.

```
URL = "HTTPS://OFFICIAL-JOKE-  
API.APPSPOT.COM/RANDOM_JOKE"
```





GET & PRINT THE JOKE!

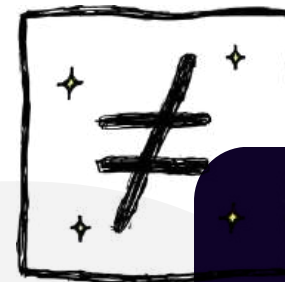
COMPLETE CODE TO FETCH YOUR JOKE

```
response = requests.get(url)
```

This sends your request to the API!

Python asks the joke website for a random joke. The response holds everything the API sends back to us.

Now we can read it!



GET THE JOKE AND PRINT IT!



```
data = response.json()
This reads the joke into Python!
print(data['setup'])
Shows the joke question
print(data['punchline'])
Shows the funny answer!
```

NOW LET'S SEE THE MAGIC HAPPEN! 🎉





YOUR TURN!

TRY THE JOKE API

NOW IT'S YOUR TURN TO CODE!

You've learned how Python talks to APIs. Now try it yourself! Open your code editor, type the joke code we learned, and run it. Can you get a different joke each time? Try running it 3 times and see what happens! 🚀



CHANGE THE CODE OR RUN IT YOURSELF



CAN YOU GET A NEW JOKE EACH TIME?



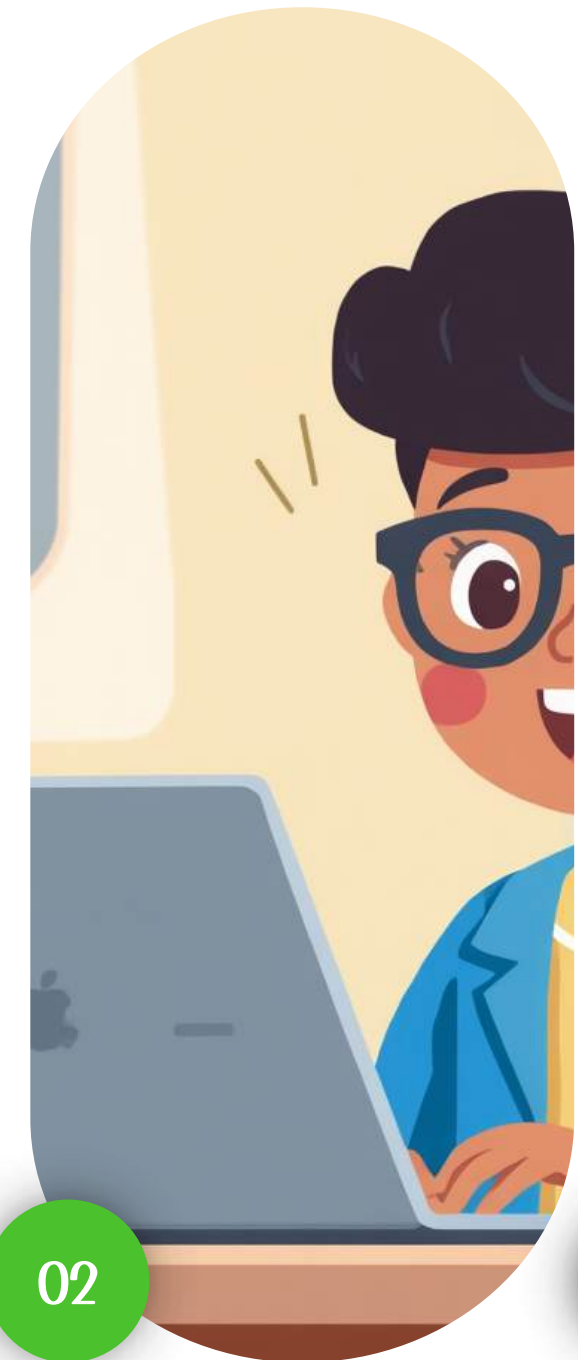
ACTIVITY TIME! 🎮



**YOU JUST MADE
PYTHON TALK TO
THE INTERNET -
AMAZING! 🤖**

KEEP GOING!

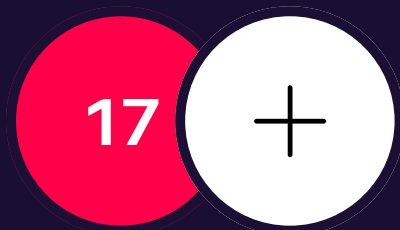
01



02

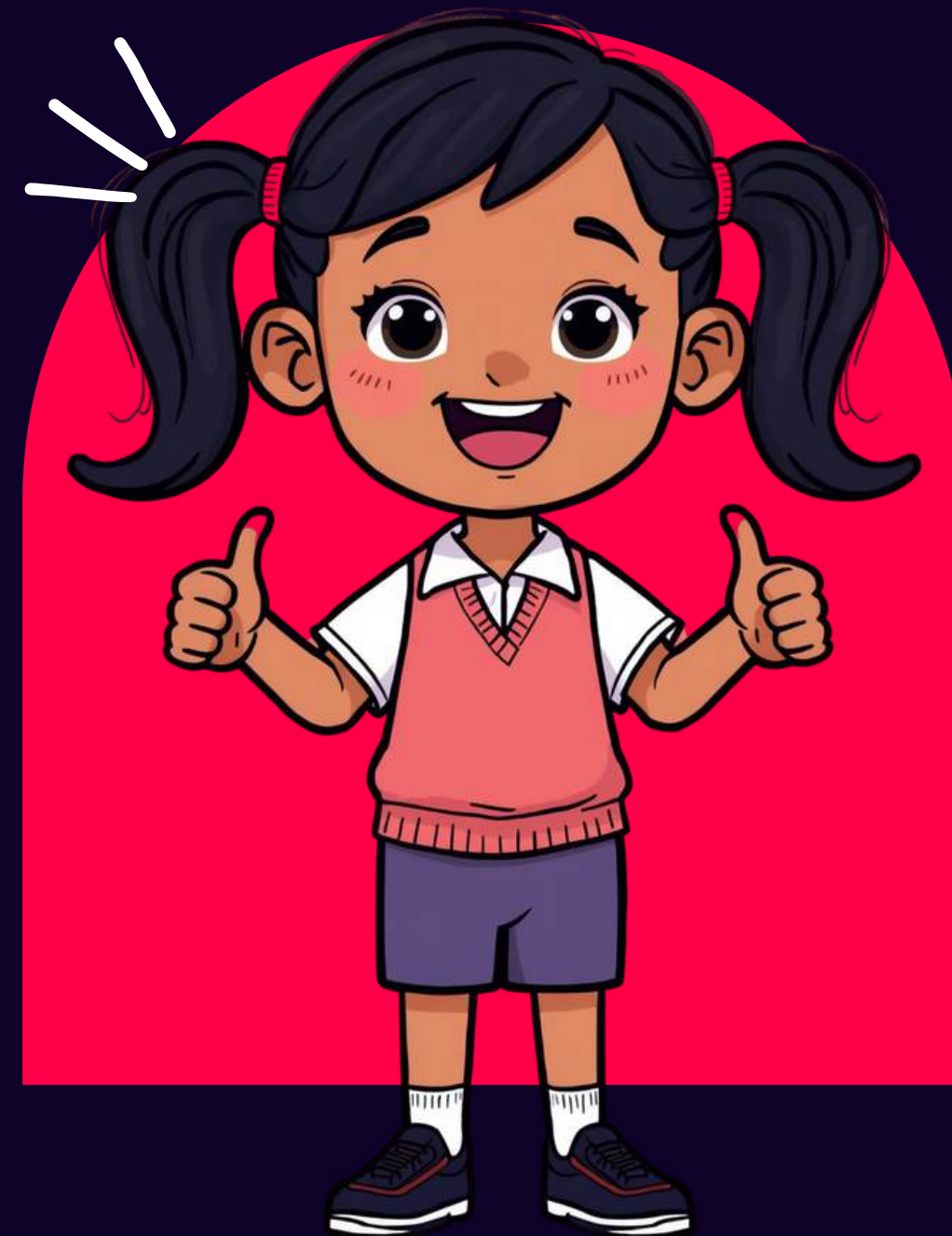


03



SUMMARY

WHAT WE LEARNED!



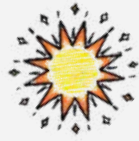
API = WAITER WHO CARRIES REQUESTS AND
RESPONSES BETWEEN YOU AND THE SERVER

REQUEST = ASKING FOR DATA, RESPONSE =
GETTING DATA BACK FROM THE API

JSON = DATA FORMAT LIKE AADHAAR CARD,
REQUESTS = PYTHON'S API TOOL



API VOCABULARY FLASHCARDS



**API: WATER BETWEEN
YOU AND DATA**



**REQUEST: ASKING FOR
SOMETHING**



**RESPONSE: ANSWER
FROM API**



VOCABULARY TIME!

JSON: Data format that computers share with each other, like a dictionary!

requests library: Python's special tool to talk to APIs and get data from the internet.

URL/Endpoint: The API's address on the internet, like a house address!

Status code: A number message telling if your request worked (200 = success, 404 = not found).



EXIT TICKET



We will share some of your creative answers next time! Get ready to hear what your classmates came up with too! 🌟



EXPLAIN AN API TO A FRIEND USING YOUR OWN ANALOGY

Write or say your answer in your own words. Think of a fun example from your daily life - like ordering food, asking a librarian for a book, or sending a message to a friend!

SHARE NEXT SESSION!





NEXT SESSION

WEATHER API FUN! ☀️🌧️



Next time, we'll use a REAL weather API! You'll write Python code to check if it will rain in your city tomorrow. How cool is that?



Get ready to become a weather detective with Python! We'll fetch live weather data from the internet. Can't wait to see you then! 🌈☀️🌧️



COMING UP NEXT!



YOU DID IT!



You just learned how Python talks to the internet! Keep practicing, you're doing great !! See you next time! 🌟

